AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

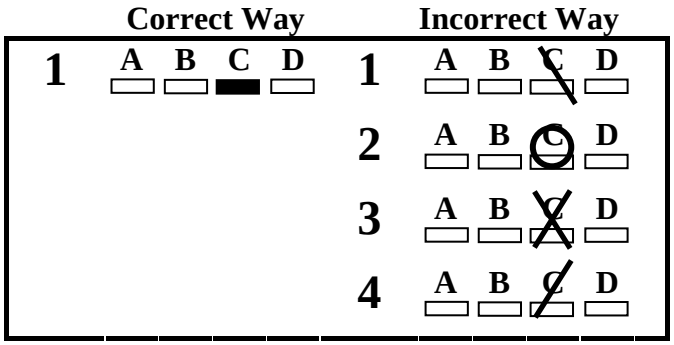
CLASS IX EXAMINATION 2010

Physics Paper I

Time allowed: 35 minutes Marks 25

INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 25 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the box for your choice with a pencil as shown below.



5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new box.
6. DO NOT write anything in the answer grid. The computer only records what is in the boxes.
7. You may use a simple calculator if you wish.

1. The bottle in the given figure contains 5 litres of vegetable oil. The word 'litre' is a unit of



A. mass.
B. weight.
C. volume.
D. density.

2. The number of significant figures in 0.093827 is

A. two.
B. five.
C. six.
D. seven.

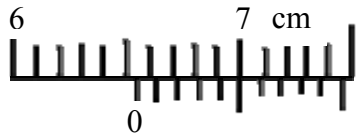
3. A metre rod has a diameter of 4.75 cm. Its value in metres is

A. $4.75\times10^{-3}\,m$
B. $4.75\times10^{-2}\,m$
C. $47.5\times10^{-2}\,m$
D. $457\times10^{-3}\,m$

4. If a resultant vector forms a 45° angle with base, then the angle between the $\vec{F}_x$ and $\vec{F}_y$ components of the vector is

A. a right angle.
B. an acute angle.
C. a reflex angle.
D. an obtuse angle.

5. The diagram shows a part of a vernier caliper. What is the reading?



A. 6.50 cm
B. 6.55 cm
C. 7.45 cm
D. 7.50 cm

6. A ball of 2 kg takes 2 sec to reach the ground from a height of 100 m. What time will ball of 4 kg take if it is falling from the same height?

A. 2 sec.
B. 4 sec.
C. 8 sec.
D. 16 sec.

7. All of the following effects are due to the application of force EXCEPT

A. recoil of a gun.
B. acceleration of a car.
C. turning of a driving wheel.
D. thermal expansion of a rod.

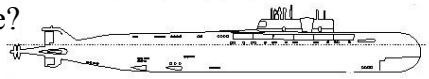
8. Which one of the following involves least friction?

A. Walking along a road.
B. Ski-ing down a snow slope.
C. Leaning a ladder against a wall.
D. brake blocks for a bicycle.

9. If a force of 20 N acts on a body, then its y-component is

A. 0
B. 1 N
C. 10 N
D. 20 N

10. A submarine sinks into water with a uniform speed of 2 m/s. What is the resultant force exerted on the submarine?



A. 0
B. 500 N
C. 2000 N
D. 10000 N

11. For greater stability a racing car should have

A. a rear engine.
B. a small chassis.
C. an oblong body.
D. a wide wheelbase.

12. The value of **g** at the Earth's poles is less than that at the equator because the distance between the

A. poles and equator is smaller.
B. poles and equator is greater.
C. poles and centre of the Earth is smaller.
D. equator and centre of the Earth is smaller.

13. The Earth and planets move around the Sun, due to a centripetal force which is provided by the gravitational force of the

A. Sun.
B. Earth.
C. Moon.
D. Planets.

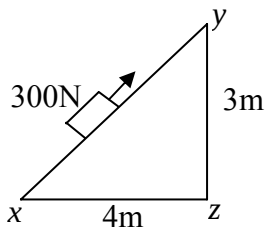
14. As the distance between the Earth and a body increases, the value of **g**

A. decreases.
B. becomes zero.
C. remains constant.
D. varies in an unpredictable manner.

15. The energy stored in a stretched string is

A. kinetic energy.
B. chemical energy.
C. elastic potential energy.
D. gravitational potential energy.

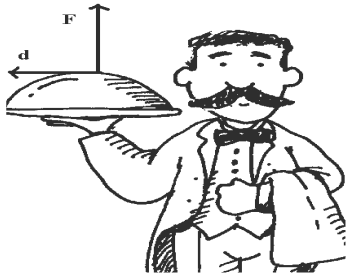
16.



A 300 N force is applied to a box in the direction of **y** in order to move it up. How much work is done?

A. 900 J
B. 1200 J
C. 1500 J
D. 30000 J

17. In the given picture calculate the work done by the waiter.

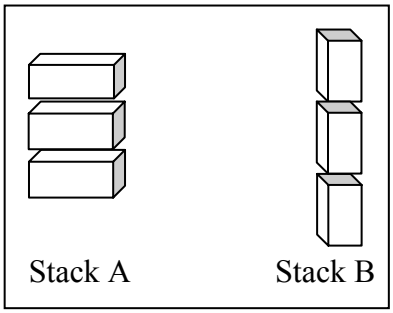


A. W = 0
B. W = F × d
C. W = –F × d
D. W = F × d Cosθ

18. A small box weighing 20 N and having an area of 0.1 m^2 stands on a floor. What is the pressure applied by the box to the floor?

A. 2.0 N/m^2
B. 19.9 N/m^2
C. 20.1 N/m^2
D. 200 N/m^2

19.



Which statement about the above figures is correct?

A. The force of stack A is greater than stack B.
B. The force of stack B is greater than stack A.
C. The pressure of stack A is greater than stack B.
D. The pressure of stack B is greater than stack A.

20. A drinking straw, a syringe and a vacuum cleaner all work on

A. Pascal's law.
B. pressure law.
C. Archimedes' principle.
D. atmospheric pressure.

21. If 4 kg of water cools from 80°C to 30°C, it loses

A. 840 J
B. 8400 J
C. 84000 J
D. 840000 J

22. In a clinical thermometer, mercury does not fall back to the bulb because

A. it is in a capillary tube.
B. it is small in quantity.
C. of the shape of the thermometer.
D. of the constriction in the tube.

23. A thermos flask has inner silver walls to reduce loss of heat by

A. radiation.
B. conduction.
C. convection.
D. evaporation.

24. All of the following are methods to prevent heat loss from houses in winter EXCEPT

A. tiled flooring.
B. carpeted floor.
C. double ceiling.
D. double glazed windows.

25. All of the following are included in the process of heat transfer EXCEPT

A. filtration.
B. radiation.
C. convection.
D. conduction.

END OF PAPER